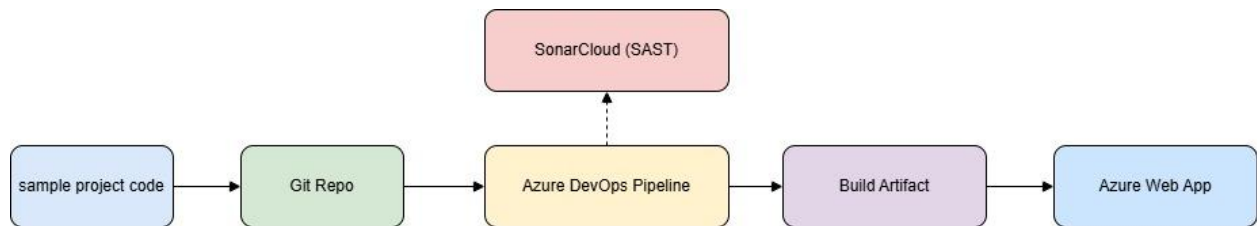


Project: Secure CI/CD Pipeline using Azure DevOps

Objective:

To build an CI/CD pipeline integrating security tools like SonarCloud to ensure code quality and vulnerability detection.

Architecture diagram:



Tools & Technologies:

Category	Tool
CI/CD	Azure DevOps
Code Quality	SonarCloud
Language	.NET
Cloud	Azure
Repo	Azure Repos

Project Setup flow:

- Setup azure devops project: Create new project in Azure DevOps and Import any official sample code from github to azure devops repo.
- Import any official sample code from github to azure devops repo. In my project I am using this
- Create 3 branches: dev, release and main.
- For **Security Implementation**:
 - Integrated SonarCloud for static code analysis. Ensured no critical vulnerabilities before deployment.
 - Setup sonarcloud: Create project and get **Project Key, Organization and Token**.
 - create Service Connections: Azure subscription and SonarCloud.

- **Create CI((Build + Scan) and CD(Deployment) Pipeline using YAML script.**

Pipeline Design:

Build Stage

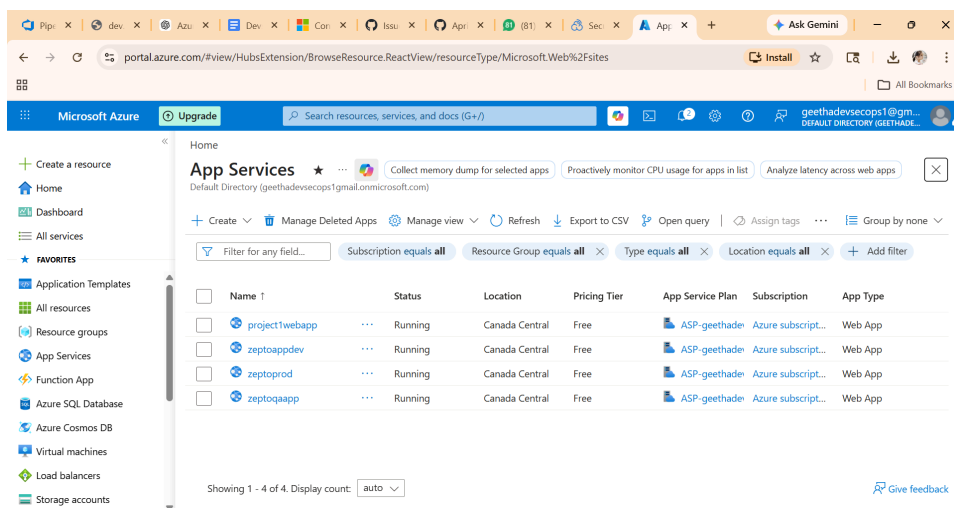
- Restore dependencies
- Build solution
- Run tests

♦ Security Stage

- SonarCloud analysis
- Code quality checks

♦ Deploy Stage

- Deploy to Azure Web Apps using AzureRmWebAppDeployment task. Different environments are triggered based on branch conditions.
 - Dev environment → **zeptoappdev**
 - QA environment → **zeptoqaapp**
 - Prod environment → **zeptoprodapp**



YAML Pipeline Explanation:

Trigger:

Pipeline runs on dev, main, and release branches.

Stages:

1. Build – compile and test code
 2. Deploy – deploy to dev environment
 3. QA – deploy to QA based on branch condition
 4. Prod – deploy only from release branch
- **Validate Pipeline:** check Build Success, Sonarcloud report, quality gate passed, deployment success

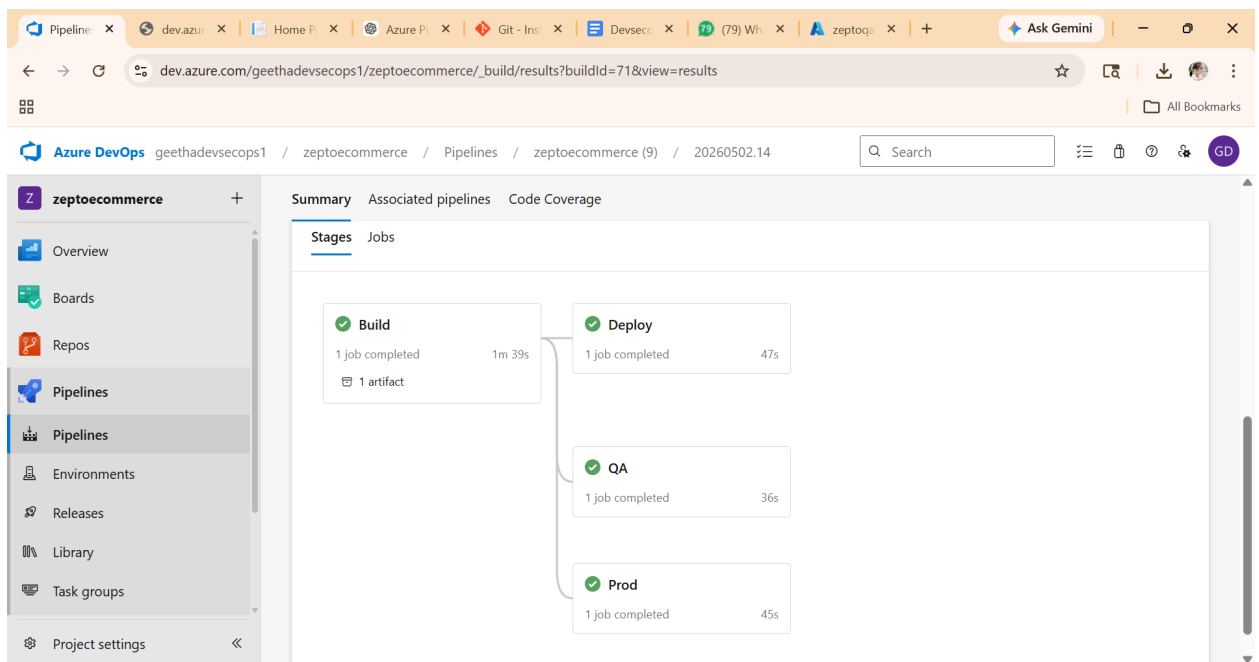


Fig: Pipeline success

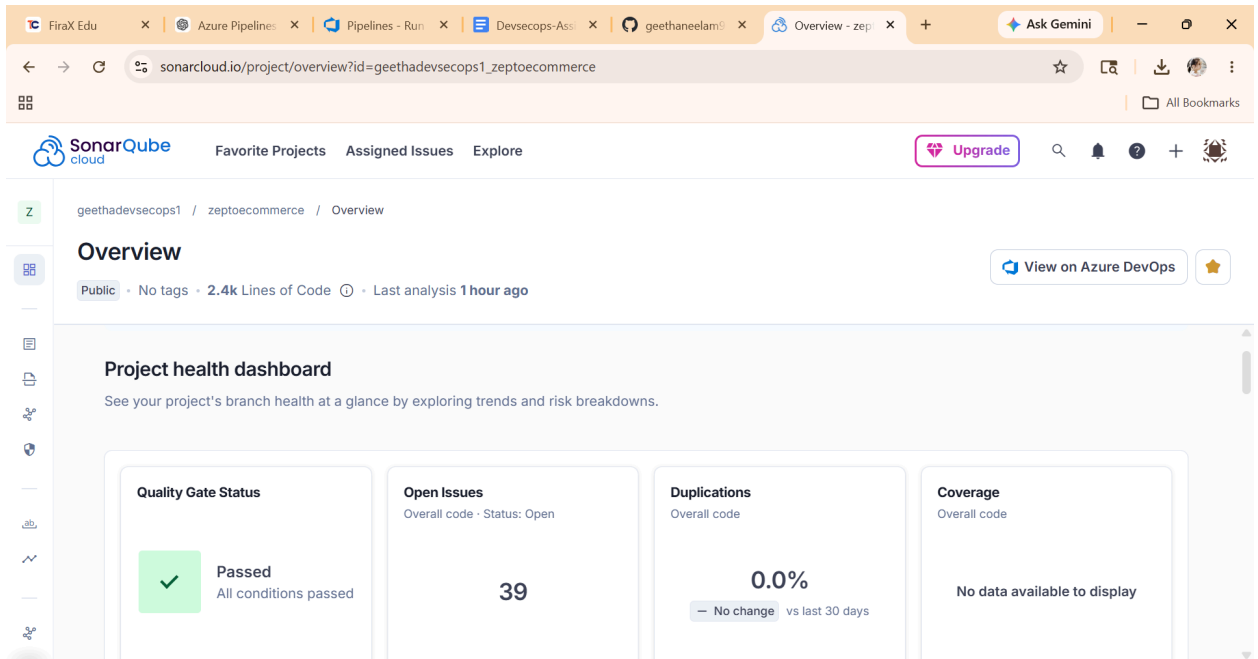


Fig:sonarqube cloud report

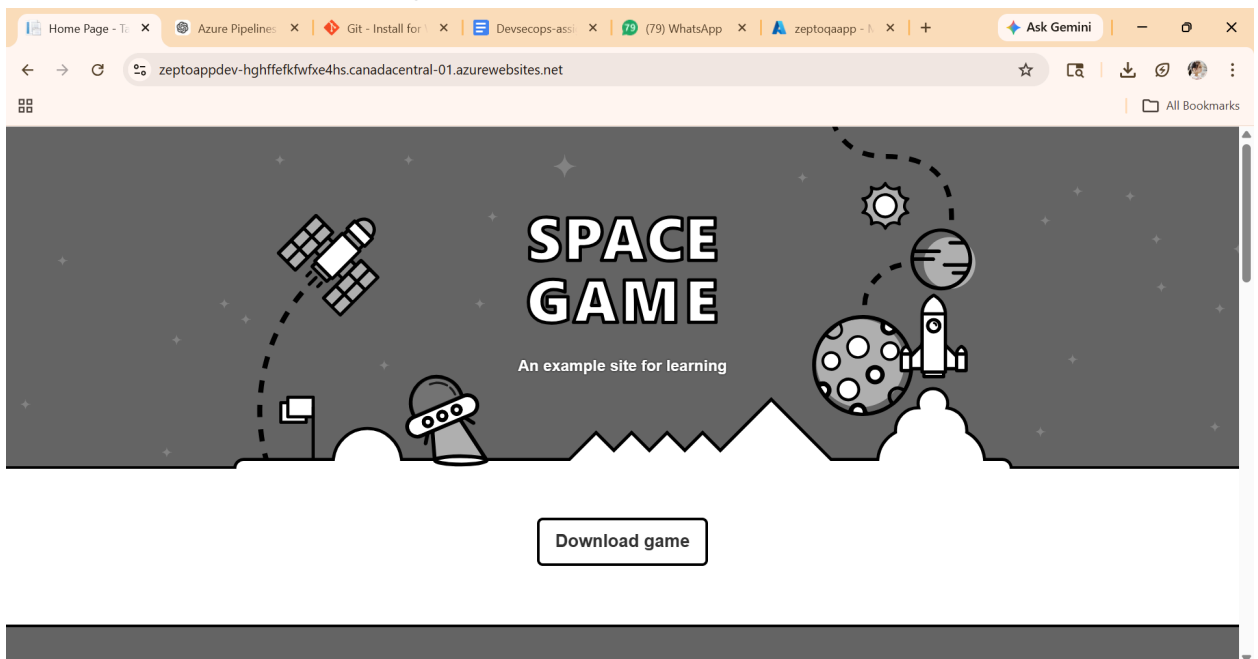


Fig:deployed on dev environment

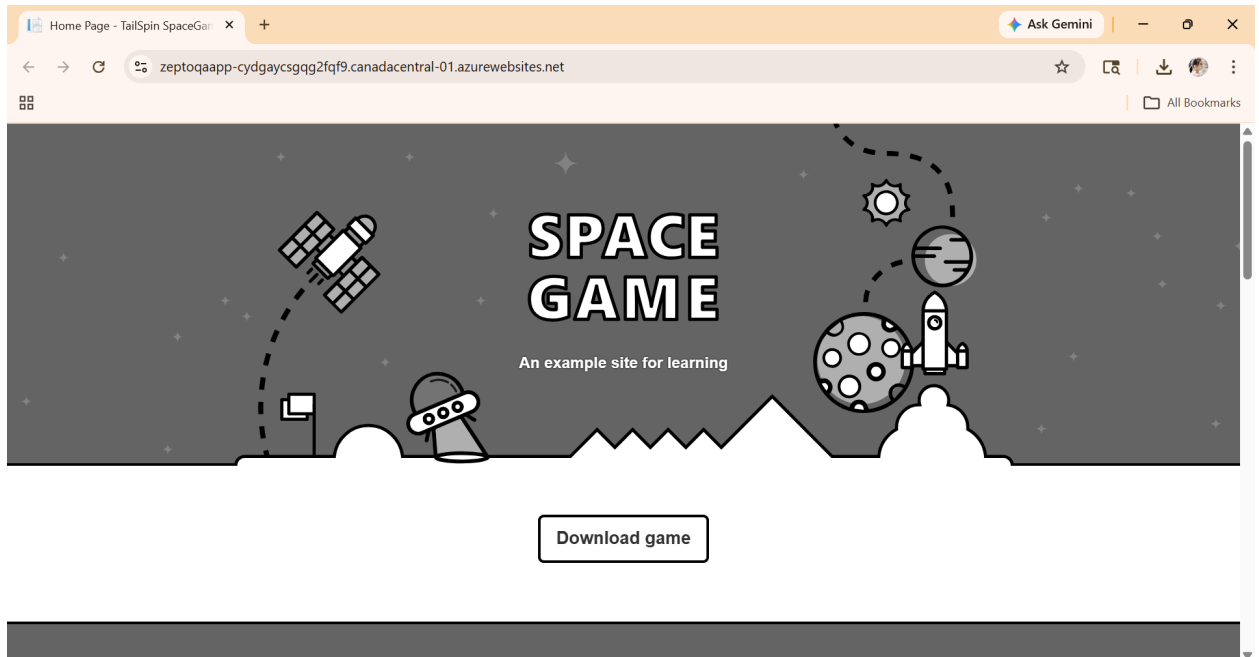


Fig:Deployed on QA environment

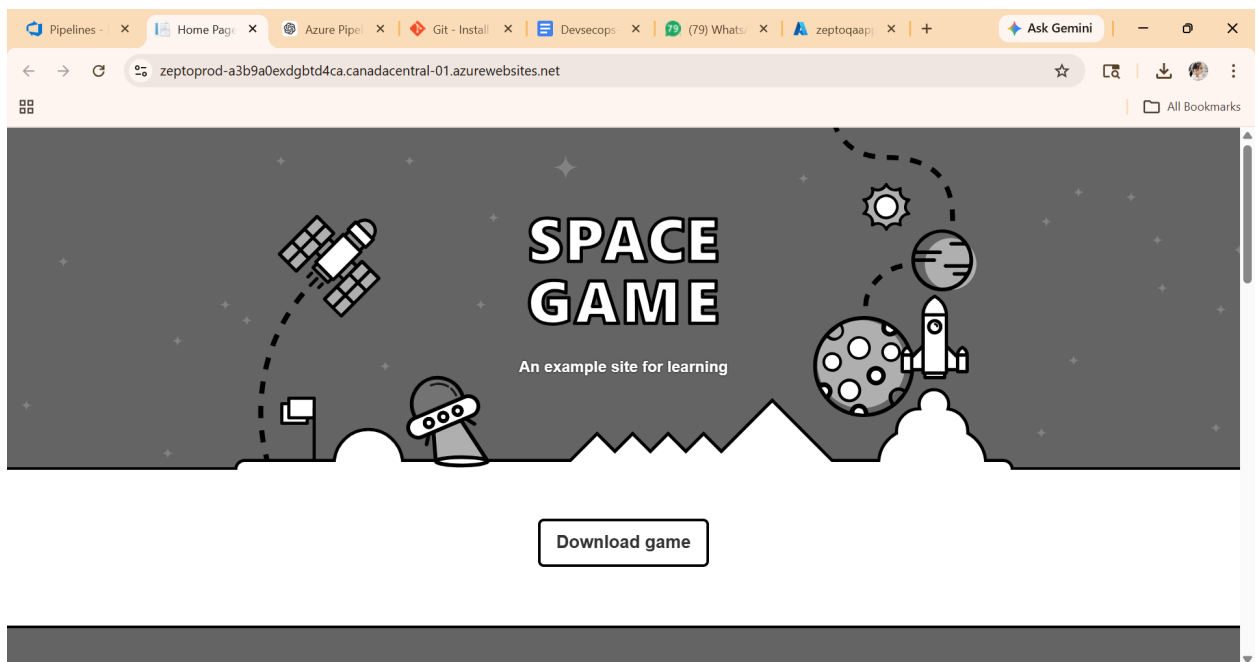


Fig:deployed on Prod environment

